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Batch – S11

Roll no. – 07

Aim – Write a program to double a number using both parameter passing techniques and analyse the results.

Code:

```
#include<stdio.h>

void passbyvalue(int a)
{
    a = 2*a;
    printf("\n Function name:Pass by Value \n Variable name: x \n Variable value: %d\n Variable address:%p \n\n",a,&a );
    return a;
}
void passbyreference(int *a)
{
    *a= *a*2;
    printf("\n Function name:Pass by Reference \n Variable name: x \n Variable value: %d\n Variable address: %p \n\n",*a,&a);
    return a;
}
void main()
{
    int a=10;
    printf("\n Function name:Main\n Variable name: a \n Variable value: %d\n Variable address%p \n\n",a,&a);
    passbyreference(&a);
    printf("\n Function name:Main\n Variable name: a \n Variable value: %d\n Variable address%p \n\n",a,&a);
    passbyvalue(a);
    printf("\n Function name:Main\n Variable name: a \n Variable value: %d\n Variable address%p \n\n",a,&a);
}
```

Output:

Function name:Main

Variable name: a

Variable value: 10

Variable address0x7ffd7ed5dbfc

Function name:Pass by Reference

Variable name: x

Variable value: 20

Variable address: 0x7ffd7ed5dbd8

Function name:Main

Variable name: a

Variable value: 20Variable address0x7ffd7ed5dbfc

Function name:Pass by Value

Variable name: x

Variable value: 40

Variable address:0x7ffd7ed5dbdc

Function name:Main

Variable name: a

Variable value: 20

Variable address0x7ffd7ed5dbfc

Working of a Stack:

Stack during call for function pass by value	Stack during call for function pass by reference
Local variable (a) of callee function of pass by value has value = 40 and address : 0x7ffffe19f98c	Local variable , (*a) of callee function of pass by reference has value = 20 and address : 0x7ffffe19f988
Passed parameter (a) of caller function Main has address 0x7ffffe19f9a4	Passed parameter (a) of caller function Main has address 0x7ffffe19f9a4